



Modular chips for killers?

DARPA wants to use modular chips for its Killer robots and drone swarm
digit.in/DarpaKillers



Protect Astronaut's health

NASA is trying new experiments to protect astronaut's health in space
digit.in/NASAexperiments

From the labs

that inform the body which microbes are the enemy. After all, our body hosts a huge number of microorganisms, and thus this war is waged constantly. Most of the foreign microbes, however, aren't harmful, and are even necessary in some cases. In fact, there are studies that have shown that our mood is greatly affected by the bacteria in our gut, which seems to turn our conception of a mind on its head. Not all these visitors have good intentions though, and some can be downright deadly. So, we made a weapon - Antibiotics.

Antibiotics - The microbial machine guns

Our immune system registers antigens much like a Pokedex registers pokemon. Oftentimes, we are infected by a microbe that our body has never encountered, and sometimes, the microbe starts to win the war. Yet, to survive, we can go to a doctor, who prescribes us medicine to ingest. Now the weapons join the war and change the tide, killing the foreign cells but not our human cells. Of course, in this war too there are casualties, but the friendly bacteria return soon enough. To understand the enemy boss, the superbug, we need to understand how the war is fought and how the antibiotics work.

The name of the war game is Eradicate the Enemy, and this is achieved by hampering essential processes in the enemy's life-cycle. This means that antibiotics work by either damaging the bacteria's DNA and preventing the repair, inhibiting the production of some crucial substance like folic acid, or weakening the cell wall until the bacteria bursts open like an over-stuffed water balloon. These drug actions don't disturb human cells because of key physiological differences. For example, Penicillin works by preventing peptidoglycan (a macromolecule that human cells don't need because we don't have cell walls) from crosslinking. Peptidoglycan are like the bricks for the bacteria's cell walls, so without crosslinking they are as useful as a brick wall without cement. Another antibiotic, Tetracycline, works by inhibiting protein synthesis. In human cells, as well as bacterial cells, ribosomes are responsible for building proteins. Ribosomes take a strand of mRNA which is like

the recipe, and then float around the cell picking up amino acids (the ingredients) and building up the protein. Tetracycline binds to a specific site on the ribosome where it is morphologically similar to, but chemically different from, the ingredient that it is replacing. This makes it impossible for the ribosome to finish up the proteins it is building, and thus the cell soon dies. Luckily it doesn't affect us because in human cells, tetracycline doesn't accumulate enough to cause this blocking of protein synthesis. The overarching theme of antibiotic action is in its name - it selec-



They're even as colorful as candy!

tively hampers a biological process. This, however, also means that the hampering can itself be hampered, by another molecule or enzyme that blocks the blocking action of the antibiotic.

The Super Nano-villain

The problem with a life form fighting a life form is that life forms adapt and evolve. In this case, bacteria have a much shorter life cycle than us, and go through thousands of generations in a short (human) time. Therefore they can evolve much faster, passing down desirable traits from parent to progeny. Add to this the bonus of horizontal/lateral gene transfer, and things start to look scary. Horizontal gene transfer makes it possible to transmit genetic information (and the corresponding trait) from one organism to another, within the same

generation. In fact, horizontal gene transfer can even take place among bacteria of different species! The more they are exposed to some environment, the faster they can adapt to it. In this way, our weapons only serve to make the enemy stronger!

This finally brings us to the superbug, the micron sized enemy boss, and guess what? It is real, and it's almost here. 'Superbug' is a term used to describe strains of bacteria that are resistant to the best antibiotics used to treat them. In other words, they're really tough little genetically enhanced micro-critters. This does not mean that an infection by a superbug strain will be any more severe than a normal infection. However, the infection will not be suppressed easily, and may eventually prove to be fatal. Some of the various strains of superbugs that have been identified till date are Methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant *Enterococci* (VRE), carbapenem resistant *Klebsiella pneumoniae*, *Clostridium difficile*, Extensively Drug Resistant Tuberculosis, Shiga toxin-producing *E. coli*, and most recently Colistin resistant *E. coli*. In fact, one of the first superbug genes was identified in New Delhi, NDM1 (New Delhi metallo-beta lactamase 1) and is now found across the world. While many species of bacteria across the world develop resistance to the most common antibiotics, our human newborns depend on the same antibiotics for survival in early life. This has been leading to an increasing number of child deaths from multi-drug resistant infections. In the last three years, superbugs have claimed more than 1,50,000 infant lives in India alone.

The increasing number of pharmaceutical companies and factories in India has also been ironically detrimental to our nation's health. The recent antibiotic boom has led to declining prices. With many drugs being available over the counter for self-medication, and many doctors easily prescribing antibiotics for almost any little infection, antibiotics are currently more common than Hajmola candy. The problems with this are manifold. Antibiotics don't work against viral infections like the flu or common cold, and specific infections may require specific antibiotics.